IME 775 — Lecture 22

Latent Spaces, PCA, and Autoencoders

Chapter 14 of Math & Architectures of Deep Learning — Part 1 of 2

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1. Motivation: The Space of Natural Signals

A digital image of height H and width W with 24-bit RGB colour is, in principle, a point in a space of size $2^{24\,HW}$. For a 28×28 MNIST digit alone that is already $2^{24 \cdot 784} \approx 10^{5664}$ configurations — a number so vast that storing even one bit per image would require more matter than exists in the observable universe.

Yet the images we actually care about — natural photographs, handwritten digits, faces of real people, paragraphs of English prose — occupy only a **vanishingly small fraction** of that ambient space. Two observations explain why:

1. **Local correlation.** In a natural image, neighbouring pixels have similar colour. Random images do not. Therefore natural images live in a highly correlated, highly constrained subset of the ambient space.
2. **Low-entropy distributions.** When we further restrict attention to images that share some property (all contain giraffes; are all sevens; are all photographs of one particular human face), the valid points form tight **clusters** inside that already-constrained subset. In stochastic language, the distribution $p(\vec{x} \mid \text{class})$ is low-entropy, highly non-uniform.

These clusters are not scattered blobs. They tend to arrange themselves **along low-dimensional surfaces** — **manifolds** — embedded inside the high-dimensional ambient space.

> **The central idea of Chapter 14.** The data of interest concentrates on (or near) a lower-dimensional manifold in the ambient input space. A good representation of the data throws away the ambient directions that are *off-manifold* (noise, background, irrelevant variation) and keeps only the *on-manifold* coordinates (the ones that index the structure we actually care about). That compressed representation is called the **latent vector** or **embedding**.

This chapter develops three progressively powerful techniques for discovering and using such manifolds:

Technique	Manifold shape	Training signal
PCA (§4)	best-fit **hyperplane**	variance maximization
Autoencoder (§6)	arbitrary **curved hypersurface**	reconstruction loss
Variational Autoencoder (Part 2)	curved hypersurface + **smooth, regularized** parameterization	reconstruction + KL divergence

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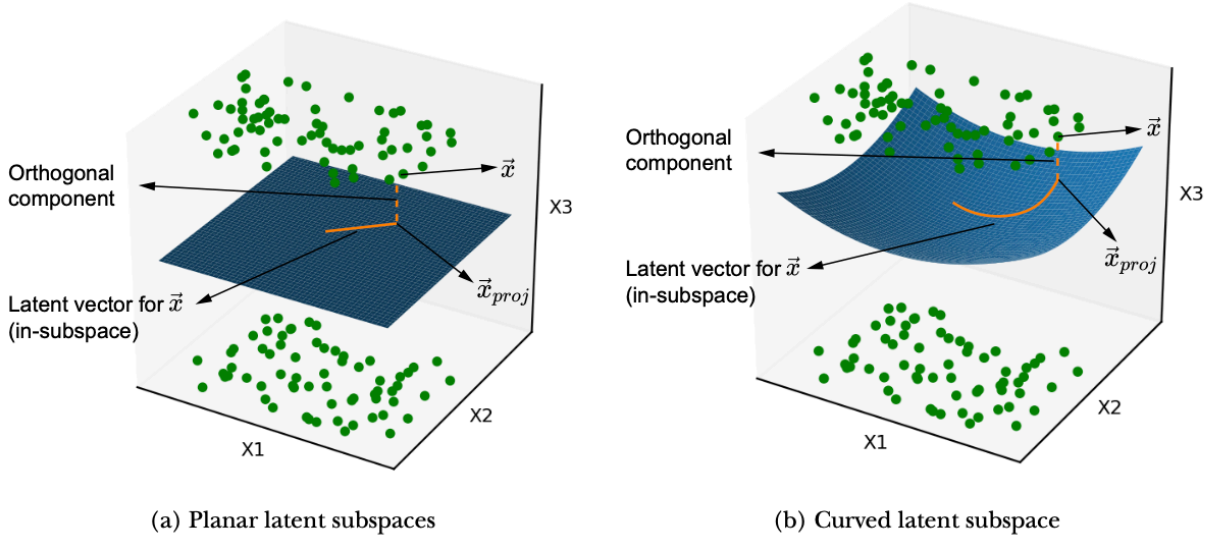
2. Geometric View of Latent Spaces

Consider the mental picture of figure 14.1 in the text. A cloud of data points (each a high-dimensional vector) hovers around some surface. Two canonical sub-cases:

Case (a) — Planar manifold. The points cluster around a flat hyperplane. A single normal direction points *off* the plane, and two (or more) in-plane directions parameterize travel *along* the plane.

Case (b) — Curved manifold. The points cluster around a curved surface (think of a swiss roll or an S-curve). At each point on the surface we can still locally define "in-manifold" directions and "off-manifold" directions, but the in-manifold directions rotate as we move around.



(a) Planar latent subspaces (b) Curved latent subspace

Figure 14.1 Two examples of latent subspaces, with planar and curved manifolds, respectively. The solid line shows the latent vector, and the dashed line represents the information lost by projecting onto the latent subspace.

2.1 Decomposing the Input Vector

Pick a training point $\vec{x}$. Project it onto the manifold to obtain the **in-manifold component** $\vec{x}_{\parallel}$. The leftover $\vec{x}_{\perp} = \vec{x} - \vec{x}_{\parallel}$ is **orthogonal** to the manifold. Any input vector decomposes as

$$\vec{x} = \vec{x}_{\parallel} + \vec{x}_{\perp}, \qquad \vec{x}_{\perp} \perp \text{manifold at } \vec{x}_{\parallel}.$$

The **latent vector** $\vec{z}$ is the representation of $\vec{x}_{\parallel}$ in some coordinate system *intrinsic* to the manifold (e.g., arc-length along a curve, or (u, v) coordinates on a surface). Crucially, $\vec{z}$ lives in a space of **smaller dimension** than $\vec{x}$ — if the manifold is k-dimensional, $\vec{z} \in \mathbb{R}^k$ even though $\vec{x} \in \mathbb{R}^d$ with $k < d$.

2.2 Why Discarding $\vec{x}_{\perp}$ Is OK (and Often Helpful)

Under the working assumption that the manifold captures *everything that matters* for the task (giraffeness, digit-class, sentiment, ...), the orthogonal part $\vec{x}_{\perp}$ is by construction the variation *unrelated* to the task — background pixels, sensor noise, lighting artefacts, stylistic jitter. Throwing it away:

- **Reduces dimensionality** (smaller, faster representations).
- **Denoises** the data (orthogonal component is noise by assumption).
- **Compresses** with minimal task-relevant information loss.
- **Reveals similarity** (two images that map to nearby $\vec{z}$ are semantically similar even if their pixel-level difference is large).

2.3 Projection Is Irreversible

A subtle but important point. The latent vector $\vec{z}$ records the **position on** the manifold, but not the **distance from** the manifold. If we try to reconstruct $\vec{x}$ from $\vec{z}$ alone we can only recover $\vec{x}_{\parallel}$, not $\vec{x}$ itself. The decoder remembers, *on average*, where the manifold sits in the ambient space, which is enough to invert the encoder approximately — but the original orthogonal offset is **irrevocably lost**.

> Reconstruction is approximate. **Exactness is traded for compression.**

2.4 Inferencing via Distance-to-Manifold

At test time, given an arbitrary new input $\vec{x}_{\text{new}}$, we can compute its distance from the learned manifold. A small distance says "this point looks like something in the training distribution"; a large distance says "this point is off-manifold — probably not a giraffe". This is how latent-space models produce **probabilistic** answers rather than hard classifications.

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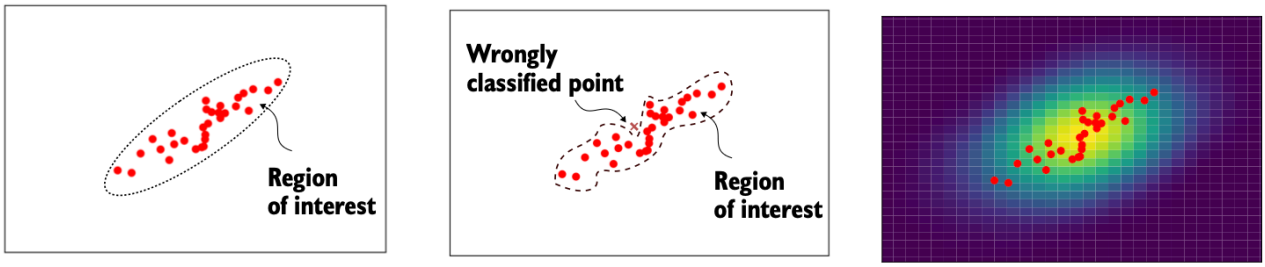
3. Generative vs. Discriminative Classifiers

Latent-space models are naturally **generative**. Understanding the contrast with **discriminative** models clarifies their advantages.

3.1 Definitions

	Discriminative	Generative
Models	$p(\text{class} \mid \vec{x})$ — decision boundary	$p(\vec{x} \mid \text{class})$ and/or $p(\vec{x})$ — density of the class
Output at test time	hard label (plus optional bounding box)	**probability** of belonging to a class
Failure mode	can overfit, adjusting the boundary to follow nooks and bends of the training data	bounded by a smooth density — **structurally resistant** to that kind of overfit
Can we sample new class instances?	no	**yes**
Convert to the other?	⟵ threshold a generative model's probability	not directly

The text's figures (14.2a–c) compare a smooth discriminative boundary, a zig-zag overfit discriminative boundary, and a generative heat-map. The heat-map elegantly avoids the question "which side of the boundary?" by assigning every point in the input space a probability of belonging to the class of interest.



(a) A good discriminative classifier—smooth decision boundary

(b) A bad discriminative classifier—irregular decision boundary

(c) Generative model—no decision boundary (heat map indicates the probability density)

Figure 14.2 Solid circles indicate training data points (all belonging to the class of interest). The dashed curve indicates the decision boundary separating the class of interest from the class of non-interest. In a generative model, there is no decision boundary. Every point in the space is associated with a probability of belonging to the class of interest (indicated as a heat map in figure 14.2c).

3.2 Three Concrete Advantages of Generative Models

1. Smoother decision surfaces. Because a generative model parameterizes a smooth probability density, it cannot suddenly carve out an irregular protrusion to capture a single outlier. This is a form of **inductive bias**: when training data is scarce, generative models generalize better.

Why the boundary is smoother

A discriminative model learns *where the boundary is* — it has no obligation to make it smooth. Given enough parameters it will bend and twist the boundary to correctly label every training point, including outliers.

A generative model instead learns a *density* — the shape of the probability "hill" for each class. The decision boundary is a side effect of those densities, not the primary object being learned.

Concrete picture. Suppose class A fills a tight cluster with one stray outlier far away.

```
Discriminative model:        Generative model:
  A A A  |  B                  A A A    B
  A A    |  B B    ←arm        A A      B B
  A A A  |  B                  A A A    B
       ↑ boundary bends           smooth boundary
       out to grab outlier        ignores the outlier
```

The discriminative boundary grows an arm to capture that one point. On new data that arm is pure noise — it will misclassify anything that falls near it. The generative model's density hill stays smooth and centered on the main cluster; its implied boundary never sprouts the arm.

Why generative models structurally cannot overfit this way. The decision boundary is derived from Bayes' rule:

$$p(\text{class} \mid \vec{x}) \propto p(\vec{x} \mid \text{class}) \cdot p(\text{class})$$

The density $p(\vec{x} \mid \text{class})$ is constrained to a smooth parametric family. These families simply have no parameters that can create a sudden jagged protrusion. The smoothness is baked into the model's structure, not enforced through an external penalty.

Inductive bias — is it good?

Yes — inductive bias is not just good, it is mathematically necessary. Without it, learning is provably impossible.

The No Free Lunch (NFL) theorem (Wolpert, 1996):

> *Averaged over all possible problems, every learning algorithm performs equally well — including random guessing.*

Any algorithm that does better than chance on the problems *you care about* must do worse on some others. The only way to win on your target problems is to build in assumptions that favor

them. Inductive bias is the price of admission to learning anything at all.

The bias–variance tradeoff. Inductive bias trades directly against variance:

	High inductive bias	**Low inductive bias**
Model flexibility	rigid	flexible
With little data	generalizes well	overfits badly
With lots of data	may underfit	can approximate truth
Risk	wrong assumption → stuck	memorizes noise

When inductive bias hurts. Bias is harmful when the assumption is *wrong for the problem*:

- **Linear regression on a nonlinear problem** — the linearity assumption is so strong the model can never fit the data no matter how much you train. This is underfitting due to high bias.
- **CNNs on tabular data** — CNNs assume local spatial structure (nearby pixels are related). Tabular columns (age, income, zip code) have no spatial structure. The assumption is wrong, and plain MLPs or gradient boosting outperform CNNs there.
- **Naive Bayes with correlated features** — assumes features are conditionally independent. When they are correlated (e.g., number of rooms and house price), this misleads the model.

Examples across the bias spectrum:

Model	**Inductive bias baked in**
Linear regression	relationship is linear
CNN	local spatial patterns, translation invariance
RNN / LSTM	sequential / temporal structure
Transformer	pairwise attention over tokens
Naive Bayes	features are conditionally independent
Decision tree	axis-aligned splits, hierarchical structure
1-NN (k = 1)	almost none — memorizes training points exactly
Deep net, no regularization	almost none — universal approximator, fits noise too

The clearest low-bias example: 1-Nearest Neighbor. Its rule is simply "copy the label of the single closest training point." A single mislabeled outlier carves out a permanent island in the decision region. With enough data 1-NN converges to the Bayes-optimal classifier — but with

scarce data it is among the worst generalizers precisely *because* it has no assumption to fall back on.

> **Key insight.** Inductive bias is not a flaw to minimize — it is a deliberate design choice. The question is not *bias or no bias* but *is my bias well-matched to my problem?* Modern large models trend toward weaker bias (more data, larger networks) because massive datasets let the model *learn* structure rather than assume it. But even transformers bake in assumptions — attention over tokens, positional encodings, layer normalization — they just make fewer geometric assumptions than a CNN.

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2. Extra diagnostic insight. Consider a "horse detector" trained discriminatively that also calls zebras horses. You'd need extra non-horse, non-zebra images to diagnose whether the model is "useless". A generative horse detector instead reports, e.g., $p(\text{horse}) = 0.92$ for horses vs 0.68 for zebras — revealing that the model *does* distinguish them but has set an inclusive threshold.

3. New instance generation. If you have $p(\vec{x})$, you can sample. Trained on Shakespeare, the model emits Shakespeare-like paragraphs. Trained on horse photos, it emits novel horse images. This is the root of modern generative AI.

> **Rule:** Any generative classifier becomes a discriminative one by thresholding its probability output. The reverse conversion is not generally possible.

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4. Benefits and Applications of Latent-Space Modelling

Before the mathematics, a compact survey of **why** this machinery is practically valuable.

4.1 Why we do it

1. **Generative power.** All benefits of generative modelling (§3.2) apply automatically to any system built on top of a learned latent space.
2. **Attention to what matters.** Redundant information that does not help the end goal is eliminated, and the system focuses on truly discriminative features. (The canonical cartoon: mugshot recognizers trained on subjects in front of the same wall will learn to represent subject-facial-features, not the wall.)
3. **Streamlined data representation.** The latent vector is a smaller, denser version of the input with no meaningful information lost.
4. **Noise elimination.** The low-variance orthogonal-to-latent-subspace component *is* the noise, and the projection removes it.

5. **Friendlier geometry for the downstream task.** A simple but illustrative example: points inside the unit disc $x^2 + y^2 \leq 1$ vs outside cannot be separated by a straight line in (x, y), but the map $(x, y) \mapsto (r, \theta)$ turns the circle $x^2 + y^2 = 1$ into the straight line $r = 1$. A well-chosen latent space can turn nonlinear problems into linear ones.

4.2 Where it gets deployed

- **Generation of new images or text** — VAEs, GANs, diffusion models.
- **Similarity retrieval.** Distance between latent vectors captures *perceptual* similarity better than pixel distance. A white-striped-on-black shirt and a black-striped-on-white shirt look different pixel-by-pixel but should be nearby in a good latent space (they are, perceptually, the same product with inverted colours).
- **Compression.** A d-dimensional input maps to a k-dimensional latent ($k \ll d$) — a lossy but principled compressor.
- **Denoising.** Encode then decode: the orthogonal component was thrown away in encoding, so the output is de-noised.
- **Transfer learning.** Latent representations learned on a large dataset (ImageNet, LAION, all of Wikipedia) are useful initialization for downstream tasks with less data.

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5. Linear Latent Spaces: PCA Revisited

Principal Component Analysis is the simplest latent-space model: the manifold is forced to be a **hyperplane** (a planar manifold of dimension k embedded in $\mathbb{R}^d$). PCA chooses the hyperplane that passes through the data mean and maximizes the variance retained after projection.

We spend a full section on PCA because (i) it is fast and closed-form, (ii) it establishes the vocabulary for nonlinear extensions, (iii) it is the exact limit of an autoencoder with no nonlinearities, and (iv) it yields the smallest reconstruction error among all *linear* projections to a subspace of a given size.

5.1 Setup

Let the data be the matrix $X \in \mathbb{R}^{n \times d}$, one row per sample, d features. Assume for now we want to reduce to $k = d - m$ dimensions (we drop the m least-variance directions).

Step 1 — Centre the data. Compute the mean $\vec{\mu} = \frac{1}{n} \sum_{i=1}^{n} \vec{x}^{(i)}$, then overwrite each row:

$$\vec{x}^{(i)} \leftarrow \vec{x}^{(i)} - \vec{\mu}, \quad i = 1, \ldots, n.$$

We store $\vec{\mu}$ so we can un-centre later during reconstruction. From here on X denotes the **mean-subtracted** data matrix.

Step 2 — Covariance matrix. The matrix $X^\top X \in \mathbb{R}^{d\times d}$ is (up to a factor of n) the sample covariance of the features. Since it is real symmetric positive-semidefinite, it has an orthonormal eigenbasis.

Step 3 — Eigendecomposition. Write

$$X^\top X\,\vec{v}_j = \lambda_j \vec{v}_j, \qquad \lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_d \geq 0.$$

The pairs $(\lambda_j, \vec{v}_j)$ are the **principal values** and **principal vectors**. Stack the $\vec{v}_j$'s into the columns of

$$V \;=\; [\vec{v}_1 \mid \vec{v}_2 \mid \cdots \mid \vec{v}_d] \;\in\; \mathbb{R}^{d\times d}, \qquad V^\top V = I.$$

5.2 Why the Principal Vectors are the Right Directions

Suppose we project data onto a unit vector $\vec{u}$. The variance of the projected data is

$$\mathrm{Var}(X\vec{u}) \;=\; \vec{u}^\top (X^\top X)\,\vec{u}.$$

Maximising this over unit vectors $\vec{u}$ is a classic Rayleigh-quotient problem whose answer is $\vec{u} = \vec{v}_1$ (the eigenvector of largest eigenvalue) with maximum value λ_1. The next-best orthogonal direction is $\vec{v}_2$ with variance λ_2, and so on. Therefore:

> **The principal vectors are the orthogonal directions of successively maximum variance. The principal values are the variances themselves.**

If λ_j is small, the data barely varies along $\vec{v}_j$ — that direction is largely noise and can be discarded with tiny information loss.

5.3 SVD: The Preferred Algorithm

Computing $X^\top X$ and then its eigenvectors is numerically inferior to the **singular value decomposition**

$$X \;=\; U\,\Sigma\,V^\top, \qquad U \in \mathbb{R}^{n\times d},\; \Sigma = \mathrm{diag}(s_1, \ldots, s_d),\; V \in \mathbb{R}^{d\times d}.$$

The relationships are:

- $\lambda_j = s_j^2$ (principal values are squared singular values),
- The V from SVD is exactly the V of principal vectors,
- The columns of $U\Sigma$ give us the coordinates of each data point in the principal basis.

5.4 Encoding (Projection)

Drop the last m columns of V to obtain

$$V_{d-m} \;=\; [\vec{v}_1 \mid \cdots \mid \vec{v}_{d-m}] \;\in\; \mathbb{R}^{d\times(d-m)}.$$

The **latent representation** of the data is

$$\boxed{Z \;=\; X\,V_{d-m} \;\in\; \mathbb{R}^{n\times(d-m)}}$$

Each row of Z is a $(d-m)$-dimensional latent vector — the "embedding" of the corresponding input point.

5.5 Decoding (Reconstruction)

To reconstruct, pad each latent vector with m zeros on the right (so that it is now d-dimensional in the principal basis), then rotate back to the original axes using $V^\top$, then add back the mean:

$$\boxed{\tilde{X} \;=\; \big[Z \mid \mathbf{0}\big]\,V^\top \;+\; \mathbf{1}\vec{\mu}^\top.}$$

Equivalently, in SVD form, $\tilde{X} = U\Sigma_{d-m}V^\top + \text{mean}$ where Σ_{d-m} zeroes out the smallest m singular values.

5.6 Reconstruction Loss and Optimality

The squared reconstruction error is

$$\|X - \tilde{X}\|_F^2 \;=\; \sum_{j=d-m+1}^{d} \lambda_j.$$

This equals the total variance that was thrown away — the sum of the smallest m principal values. The **Eckart–Young–Mirsky theorem** guarantees that among **all** rank-$(d-m)$ linear projections, PCA achieves the minimum reconstruction error. No linear projection can do better.

5.7 PyTorch Implementation

```
import torch

x_mean = X.mean(dim=0)
Xc     = X - x_mean
U, S, Vh = torch.linalg.svd(Xc, full_matrices=False)
V        = Vh.T                          # principal vectors as columns
V_trunc  = V[:, :-1]                     # drop the lowest-variance direction

Z        = Xc @ V_trunc                  # (n, d-1) latent codes
Z_pad    = torch.cat([Z, torch.zeros(Z.shape[0], 1)], dim=1)
X_hat    = Z_pad @ V.T + x_mean          # reconstruction
```

5.8 Choosing How Many Components to Keep

Plot the cumulative variance explained

$$\mathrm{CVE}(k) \;=\; \frac{\sum_{j=1}^{k} \lambda_j}{\sum_{j=1}^{d} \lambda_j}$$

against k. The "elbow" of this curve is a common heuristic for the best latent dimensionality. Typical target: 95%–99% of variance preserved.

5.9 Worked Example: 3D Points Around the $X_0 = X_2$ Plane

Following the text's figure 14.3, suppose we generate 1,000 three-dimensional points clustered near the plane $X_0 = X_2$ with small Gaussian noise in the $X_0 - X_2$ direction.

- The covariance matrix has two large eigenvalues (variance in the plane) and one small eigenvalue (the thin noise direction normal to the plane).
- PCA with $k = 2$ throws away the normal direction, projecting every point onto the learned plane.
- Reconstruction is excellent because very little variance lived along the dropped direction.

(See the accompanying marimo notebook, §1, for a runnable PyTorch demo and 3D/2D plots.)

5.10 When PCA Fails

PCA projects to a **hyperplane**. If the data lies on a curved manifold, no plane fits well:

- **S-curve.** Data lives on a 2D curled sheet in 3D. A 2D PCA projection flattens the curl — points that were distant along the sheet get collapsed near each other.
- **Swiss roll.** Same pathology, more dramatic.
- **Sphere.** The entire data set is equidistant from every plane — reconstruction error is huge no matter which plane PCA picks.

To handle such data we need nonlinear projections. Enter autoencoders.

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6. Autoencoders

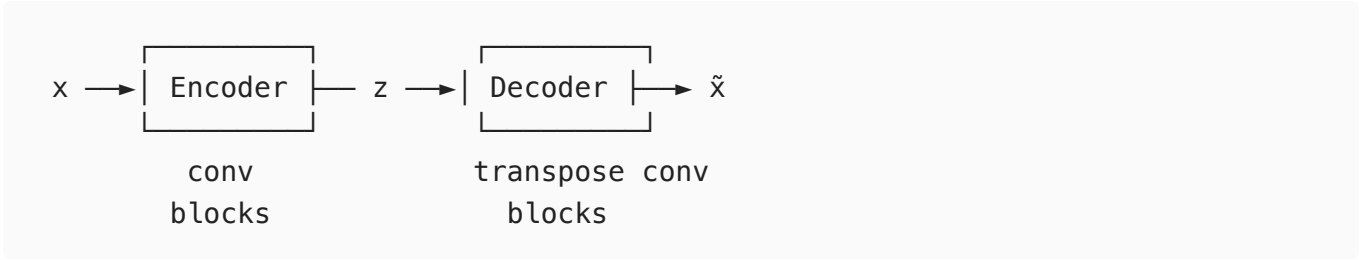
An **autoencoder (AE)** is a pair of neural networks that together learn an arbitrary nonlinear projection onto a latent manifold and its inverse.

6.1 Definition

$$\boxed{\vec{z} \;=\; E(\vec{x}), \qquad \tilde{\vec{x}} \;=\; D(\vec{z}), \qquad \mathcal{L}_{\text{recon}}(\vec{x}) \;=\; \|\vec{x} - \tilde{\vec{x}}\|^2.}$$

- **Encoder** E — a neural network that maps input $\vec{x} \in \mathbb{R}^d$ to a latent code $\vec{z} \in \mathbb{R}^{n_z}$ with $n_z \ll d$.
- **Decoder** D — a neural network that maps the latent code back to the input space, producing a reconstruction $\tilde{\vec{x}}$.
- The pair is trained **end to end** to minimize reconstruction loss.

Schematically,



The narrow point in the middle — the **bottleneck** — forces the network to compress. No task-irrelevant information can survive the squeeze.

6.2 Unsupervised Training

The "target" for an autoencoder is **the input itself**. No labels are needed. That makes autoencoders one of the canonical **self-supervised** / unsupervised methods. You can train one on every image you can scrape and then re-use the encoder as a pretrained feature extractor for downstream supervised tasks.

6.3 Convolutional Encoder for Image Data (Listing 14.2)

A standard choice for 28×28 MNIST inputs:

```
import torch.nn as nn

n_z = 8
input_image_size = (1, 28, 28)

conv_encoder = nn.Sequential(
    nn.Conv2d(1, 16, kernel_size=3, stride=1, padding=1),
    nn.BatchNorm2d(16), nn.ReLU(),
    nn.MaxPool2d(kernel_size=2),                # → 16 × 14 × 14
    nn.Conv2d(16, 32, kernel_size=3, stride=1, padding=1),
    nn.BatchNorm2d(32), nn.ReLU(),
    nn.MaxPool2d(kernel_size=2),                # → 32 × 7 × 7
    nn.Conv2d(32, 64, kernel_size=3, stride=1, padding=1),
    nn.BatchNorm2d(64), nn.ReLU(),
    nn.MaxPool2d(kernel_size=2),                # → 64 × 3 × 3
```

```
    nn.Flatten(),                              # → 576
)
fc = nn.Linear(576, n_z)                       # → n_z-dimensional latent
```

Every `MaxPool2d(2)` halves the spatial resolution, and every `Conv2d` expands the channel count. By the time we reach the linear layer the input has been squeezed from 784 dimensions down to 576 spatial features, then down to n_z.

Code walk-through

The encoder is a stack of three identical "blocks" plus a flatten + linear head. Each block does the same three things:

Block step	Purpose
`Conv2d`	learn local patterns; **expand the channel count** (more feature detectors)
`BatchNorm2d`	normalize the Conv **pre-activations** across the batch (zero mean, unit variance per channel) — stabilizes training, allows higher learning rate, and gives ReLU a clean zero-centered input so ~half the units fire
`ReLU`	nonlinearity; without it the whole stack collapses to a single linear map
`MaxPool2d(2)`	**halve the spatial resolution** by keeping the maximum in each 2×2 window

Shape evolution for one MNIST image (`1 × 28 × 28`):

```
input           : 1  × 28 × 28   (1 channel, 784 pixels)
after block 1   : 16 × 14 × 14   (Conv expands channels, MaxPool halves H,W)
after block 2   : 32 ×  7 ×  7
after block 3   : 64 ×  3 ×  3   (= 576 features after flatten)
after Flatten   : 576
after fc        : 8              ← the latent code z
```

Notice the **classic CNN trade**: spatial dimensions shrink (28 → 14 → 7 → 3) while channel depth grows (1 → 16 → 32 → 64). The network gives up "where" in exchange for "what" — losing pixel coordinates but gaining higher-level feature detectors. The final `Linear(576, 8)` is the bottleneck that forces a 98% compression (784 → 8).

Why `padding=1` with `kernel_size=3` and `stride=1` ? This is the "same padding" trick — it keeps the spatial dimensions unchanged after the convolution. All shrinking happens in the `MaxPool2d` step, so shape arithmetic stays simple.

6.4 Convolutional Decoder (Listing 14.3)

The decoder mirrors the encoder, replacing `Conv2d → MaxPool` with `ConvTranspose2d` to re-inflate the spatial resolution:

```
conv_decoder = nn.Sequential(
    # reshape n_z → 64 × 3 × 3 via a linear then view
    nn.ConvTranspose2d(64, 32, kernel_size=3, stride=2, padding=0),
    nn.BatchNorm2d(32), nn.ReLU(),              # → 32 × 7 × 7
    nn.ConvTranspose2d(32, 16, kernel_size=3, stride=2, padding=1,
output_padding=1),
    nn.BatchNorm2d(16), nn.ReLU(),              # → 16 × 14 × 14
    nn.ConvTranspose2d(16, 1,  kernel_size=3, stride=2, padding=1,
output_padding=1),
    nn.Sigmoid(),                                # → 1 × 28 × 28 in [0, 1]
)
```

Transposed convolution (Chapter 10) is the "learnable upsampling" operation. `Sigmoid` at the end squashes the output to $[0, 1]$, matching the pixel range of MNIST.

Code walk-through

The decoder runs the encoder *in reverse*. Where the encoder used `MaxPool` to **shrink** the spatial size, the decoder uses `ConvTranspose2d` (with `stride=2`) to **double** it — effectively a learnable upsampling.

Before this stack runs, the 8-dimensional latent `z` is passed through a linear layer and reshaped to `64 × 3 × 3` so it has the spatial shape the decoder expects.

Shape evolution through the decoder:

```
z              : 8                (latent code)
linear + view  : 64 ×  3 ×  3
after layer 1  : 32 ×  7 ×  7    (ConvTranspose, stride=2)
after layer 2  : 16 × 14 × 14
after layer 3  :  1 × 28 × 28    (back to original image shape)
Sigmoid        :  1 × 28 × 28    (pixel values in [0, 1])
```

Mirror image of the encoder: spatial dimensions grow (3 → 7 → 14 → 28) while channel depth shrinks (64 → 32 → 16 → 1). The network is rebuilding "where" from the compressed "what."

Why `Sigmoid` at the end? MNIST pixels live in `[0, 1]` (after normalization). `Sigmoid` squashes any real number into that range, so the output is automatically a valid image. Without it the decoder could produce pixel values like `-3.2` or `42.7`, which would inflate the MSE loss without being meaningful.

`output_padding=1` is a small bookkeeping correction that handles the ambiguity in the inverse of strided convolution — different input sizes can produce the same output size, so PyTorch needs a hint to pick the right one.

6.5 Training Loop

```
import torch.nn.functional as F

def step(x, encoder, fc, decoder, optim):
    z      = fc(encoder(x))
    x_hat  = decoder(z.view(-1, 64, 3, 3))
    loss   = F.mse_loss(x_hat, x)
    optim.zero_grad(); loss.backward(); optim.step()
    return loss
```

Typical choices: Adam optimizer, learning rate 10^{-3}, batch size 128, 10–20 epochs for a rough MNIST autoencoder.

Code walk-through

This `step` function performs **one gradient update** for a single batch of images.

The forward pass:

```
z = fc(encoder(x))
```

Two stages of encoding:

1. `encoder(x)` — runs the conv blocks, producing a flat 576-dimensional vector per image.
2. `fc(...)` — squeezes 576 → 8. `z` is the **latent code**.

```
x_hat = decoder(z.view(-1, 64, 3, 3))
```

`.view(-1, 64, 3, 3)` reshapes the flat 8-dim `z` (after a linear projection back to 576) into a 3D tensor with 64 channels and 3×3 spatial size — the format the decoder's first `ConvTranspose2d` expects. Then `decoder(...)` upsamples back to `1 × 28 × 28`. `x_hat` is the reconstructed image.

```
loss = F.mse_loss(x_hat, x)
```

Reconstruction loss — pixel-wise mean squared error between reconstruction and input. The target *is* the input (no labels needed) — this is what makes autoencoder training **self-**

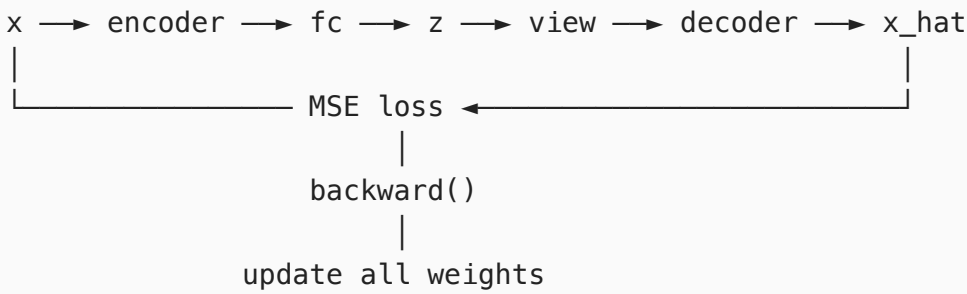
supervised.

The backward pass:

```
optim.zero_grad(); loss.backward(); optim.step()
```

Call	What it does
`optim.zero_grad()`	Clears gradients from the previous step. PyTorch *accumulates* gradients by default, so without this they would pile up.
`loss.backward()`	Backpropagates: computes $\partial \text{loss}/\partial \theta$ for every weight in encoder + fc + decoder.
`optim.step()`	Nudges every weight in the direction that reduces the loss (Adam, SGD, etc.).

Data flow:



The bottleneck (`z` is 8-dim, `x` is 784-dim) forces the network to learn a meaningful 98% compression — it cannot just copy the input through.

6.6 Asymmetric Architectures

The encoder and decoder need **not** be mirror images. It is common to use a deeper decoder than encoder when the reconstruction task is harder than the encoding task, and the reverse for fast deployment scenarios. The only constraint is that the output of the decoder has the same shape as the input.

6.7 Autoencoder vs PCA

Suppose both the encoder and decoder are single linear layers with no activation functions and no biases:

$$E(\vec{x}) \;=\; W_E\vec{x}, \qquad D(\vec{z}) \;=\; W_D\vec{z}.$$

Then the reconstruction is $\tilde{\vec{x}} = W_D W_E \vec{x}$. Minimising $\|\vec{x} - W_D W_E \vec{x}\|^2$ over rank-k matrices $W_D W_E$ recovers (up to a basis change) the PCA projection. In this limit **autoencoder = PCA**.

The instant we add a ReLU (or any nonlinearity) anywhere in the encoder/decoder, the class of realizable projections expands from hyperplanes to **arbitrary curved hypersurfaces**. This is the qualitative leap from PCA to a real autoencoder.

6.8 What Information Is Stored Where?

This is the most useful mental model:

Information about	**Lives in**
The specific input $\vec{x}$	the latent vector $\vec{z}$
The **shape and location of the manifold** in the ambient space	the decoder weights
How to **project** an arbitrary ambient point onto the manifold	the encoder weights

The decoder's weights are, collectively, an average memory of "where the data manifold sits". The encoder's weights know how to project any new point onto that manifold. Neither network knows anything about any individual training example after training — that information is carried by the latent code.

7. Limits of a Plain Autoencoder

A plain autoencoder has a serious structural weakness: **reconstruction loss does not uniquely determine the latent space**. Two radically different latent spaces can achieve equally low reconstruction error.

7.1 The Zig-Zag Pathology

Imagine 2D input data with a linear trend. We want to compress to a 1D latent space. Two candidate 1D manifolds both fit the training data:

- A **smooth straight line** that passes through the data cloud.
- A **zig-zag curve** that threads through every single training point exactly.

Both achieve **zero** reconstruction error on the training set. But they are not equally good:

	Smooth line	**Zig-zag**
Reconstruction on *training* points	0	0
Reconstruction on *held-out* points	small	huge

	Smooth line	Zig-zag
Distance between two nearby inputs **measured along the manifold**	proportional to input distance	wildly distorted
Manifold length	short	long

The zig-zag has essentially **memorised** the training set. It overfits despite having low training loss. This is analogous to a polynomial regression model with degree equal to the number of training points — perfect training fit, disastrous test behaviour.

7.2 Why This Matters

A deep, overparameterised autoencoder can easily learn a zig-zag-like latent manifold. The danger manifests in three concrete ways:

1. **Interpolation fails.** Sampling a latent vector halfway between two training codes decodes to garbage because the straight segment between them in latent space may pass through long detours in input space.
2. **Generation fails.** Sampling latent vectors from any distribution that doesn't already match the (unknown, irregular) training distribution yields garbage outputs.
3. **Similarity fails.** Euclidean distance in a zig-zag latent space has no relation to perceptual distance in input space.

7.3 What We Want: The Minimum Description Length Principle

A good latent manifold:

1. **Fits** the training data well (low reconstruction loss).
2. Is **as short / flat as possible** (few twists and turns).
3. Is **compact** (the training data's latent codes do not spread indefinitely far apart).
4. Is **continuous** — small changes in $\vec{z}$ yield small changes in $\tilde{\vec{x}}$.

Properties 2–4 echo the **Minimum Description Length (MDL) principle**: among models that fit the data, prefer the one that requires the fewest bits to describe. They also echo **Occam's Razor**: prefer the simpler of two equally predictive models.

7.4 How to Impose These Properties

Two complementary strategies:

1. **Regularization.** Add an explicit loss that penalizes undesirable latent-space structure (long manifolds, spread codes, large encoder weights).

2. **Probabilistic modelling.** Model the latent space as a probability distribution from a simple family (e.g., Gaussian) and demand that the distribution stay close to a compact target (e.g., $\mathcal{N}(0, I)$) via a divergence penalty.

Strategy 2, combined with a clever trick for keeping the network differentiable, yields **variational autoencoders** — the subject of Lecture 23.

Reference: Krishnendu Chaudhury, *Math and Architectures of Deep Learning*, Chapter 14 §§14.1–14.6.